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Batch – S11
Roll no. – 07

Aim – Implementation of Binary Search Tree

Code:

```
#include<stdio.h>

#include<stdlib.h>

struct node

{

    int key;

    struct node*left,*right;

};

struct node *newNode(int item){

    struct node *temp=(struct node*)malloc(sizeof(struct node));

    temp->key=item;

    temp->left=temp->right=NULL;

    return temp;

};

void inorder(struct node* root){

    if(root!=NULL){

        inorder(root->left);

        printf("%d->",root->key);

        inorder(root->right);

    }

}

struct node*insert(struct node*node,int key){
```

```

if(node==NULL)

return newNode(key);

if(key<node->key)

node->left=insert(node->left,key);

else

node->right=insert(node->right,key);

return node;

}

struct node*minValueNode(struct node*node){

    struct node*current=node;

    while(current && current->left!=NULL)

        current=current->left;

    return current;

}

struct node*deleteNode(struct node*root,int key){

    if(root==NULL)

        return root;

    if(key<root->key)

        root->left=deleteNode(root->left,key);

    else

        if(key>root->key)

            root->right=deleteNode(root->right,key);

        else{

            if(root->left==NULL){

                struct node* temp=root->right;

                free(root);


```

```

        return temp;
    }
    else if(root->right==NULL){
        struct node*temp=root->left;

        free(root);

        return temp;

    }

    struct node*temp=minValueNode(root->right);

    root->key=temp->key;

    root->right=deleteNode(root->right,temp->key);

}

return root;

};

int main()
{
    struct node*root=NULL;

    root=insert(root,8);

    root=insert(root,3);

    root=insert(root,1);

    root=insert(root,6);

    root=insert(root,7);

    root=insert(root,10);

    root=insert(root,14);

    root=insert(root,4);

```

```
printf("Inorder traversal:");  
  
inorder(root);  
  
printf("\nAfter deleting 10\n");  
  
root=deleteNode(root,10);  
  
printf("inorder traversal: ");  
  
inorder(root);  
  
}
```

Output:

Inorder traversal:1->3->4->6->7->8->10->14->

After deleting 10

inorder traversal: 1->3->4->6->7->8->14->

Process returned 0 (0x0) execution time : 0.031 s

Press any key to continue.